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Linear Algebra and
Differential Equations

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Assignment # (3)

Q23. $\frac{dx}{dt} = 4(x^2+1)$

$$dx = 4(x^2+1) dt$$

$$\Rightarrow \frac{1}{x^2+1} dx = 4 dt \Rightarrow \int \frac{1}{1+x^2} dx = \int 4 dt$$

$$\int \frac{1}{1+x^2} dx = \int 4 dt$$

$$\Rightarrow (\tan^{-1}(x) = 4t + C) \times \tan x$$
$$x = \tan(4t + C)$$

Given initial-value : $x = 1$; $t = \pi/4$

$$\Rightarrow x = \tan(4 \times \pi/4 + C)$$

$$1 = \tan(\pi + C)$$

$$\tan^{-1}(1) = \pi + C$$

$$\frac{\pi}{4} - \pi = C ; C = -\frac{3\pi}{4}$$

$$\therefore x = \tan(4t - 3\pi/4)$$

Q24. $\frac{dy}{dx} = \frac{y^2-1}{x^2-1} ; dy = \frac{y^2-1}{x^2-1} dx$

$$\Rightarrow \frac{1}{y^2-1} dy = \frac{1}{x^2-1} dx$$

$$\int \frac{1}{y^2-1} dy = \int \frac{1}{x^2-1} dx$$

 $\Downarrow$

$$\int \frac{1}{(y+1)(y-1)} dy \Rightarrow \frac{1}{(y+1)(y-1)} = \frac{A}{y+1} + \frac{B}{y-1} \quad \left\{ \begin{array}{l} -1 + 1 \\ 2(y+1) \quad 2(y-1) \end{array} \right.$$

$$1 = A(y-1) + B(y+1)$$

$$\Rightarrow y=1 ; 1 = B(2) \quad \left| \quad y=-1 ; 1 = A(-2) \right.$$
$$\quad \quad \quad 1/2 = B \quad \quad \quad -1/2 = A$$

$$\int \frac{-1}{2(y+1)} + \frac{1}{2(y-1)} dy \Rightarrow \int \frac{-1}{2(y+1)} dy + \int \frac{1}{2(y-1)} dy$$

$$\Rightarrow \frac{-1}{2} \int \frac{1}{y+1} dy + \frac{1}{2} \int \frac{1}{y-1} dy$$

$$\Rightarrow \frac{-1}{2} \ln|y+1| + \frac{1}{2} \ln|y-1| = \frac{1}{2} \ln|y-1| - \frac{1}{2} \ln|y+1|$$

$$(\square) a \ln A - a \ln B$$

$$\Rightarrow a \ln \left(\frac{A}{B} \right)$$

$$\Rightarrow \frac{1}{2} \ln \left(\frac{y-1}{y+1} \right)$$

$$\therefore \int \frac{1}{x^2-1} dx = \frac{1}{2} \ln \left(\frac{x-1}{x+1} \right)$$

$$\therefore \left[\frac{1}{2} \ln \left(\frac{y-1}{y+1} \right) = \frac{1}{2} \ln \left(\frac{x-1}{x+1} \right) + C \right] \times 2$$

Given the initial-value ; $x=2$; $y=2$;

$$\Rightarrow \cancel{\frac{1}{2} \ln \left(\frac{2-1}{2+1} \right)}$$

$$\ln \left(\frac{y-1}{y+1} \right) = \ln \left(\frac{x-1}{x+1} \right) + 2C$$

$$e^{\ln \left(\frac{y-1}{y+1} \right)} = e^{\ln \left(\frac{x-1}{x+1} \right) + 2C}$$

$$\frac{y-1}{y+1} = \frac{x-1}{x+1} \times (e^{2C}) \rightarrow \text{can be considered another constant, } C,$$

$$\therefore \frac{y-1}{y+1} = \left(\frac{x-1}{x+1} \right) C$$

$$(y-1)(x+1) = (y+1)(x-1)C$$

$$yx + y - x - 1 = (yx - y + x - 1)C$$

$$yx + y - yxC + yC = xC - C + x - 1$$

$$y(x+1 - xC + C) = xC - C + x - 1$$

$$\Rightarrow y = \frac{xC - C + x - 1}{x + 1 - xC + C} ; x=2, y=2 ;$$

$$\Rightarrow 2 = \frac{2C - C + 2 - 1}{2 + 1 - 2C + C} ; \frac{C+3}{3-C} = 2 \quad | \quad C+3 = 6-2C$$

$$3C = 3 \quad | \quad C = 1$$

$$y = \frac{x-1+x+1}{x+1-x+1} = y = \frac{2x}{2} \Rightarrow \underline{\underline{y=x}}$$

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Q25. $x^2 \frac{dy}{dx} = y - xy \Rightarrow x^2 dy = y(1-x) dx$

$$\Rightarrow \frac{1}{y} dy = \frac{1-x}{x^2} dx \Rightarrow \frac{1}{y} dy = \left[\frac{1}{x^2} - \frac{x}{x^2} \right] dx$$

$$\Rightarrow \int \frac{1}{y} dy = \int \frac{1}{x^2} dx - \int \frac{1}{x} dx$$

$$\Rightarrow \ln y = -\frac{1}{x} - \ln x + C$$

$$\Rightarrow e^{\ln y} = e^{-\frac{1}{x} - \ln x + C}$$

$$\Rightarrow y = e^{-1/x} \cdot e^{-\ln x} \cdot e^C$$

$$y = e^{-1/x} \left(\frac{1}{e^{\ln x}} \right) \cdot e^C; y = \frac{e^{-1/x} \cdot e^C}{x}$$

Let $e^C = C_1$ (another constant)

$$\therefore y = C \left(\frac{e^{-1/x}}{x} \right)$$

Given the initial-value: $x = -1, y = -1$,

$$\Rightarrow -1 = C \left(\frac{e}{-1} \right)$$

$$1 = Ce$$

$$1/e = C$$

$$\therefore y = \frac{1}{e} \left(\frac{e^{-1/x}}{x} \right)$$

Q26. $\frac{dy}{dt} + 2y = 1$ } first order linear differential equation

$$\frac{dy}{dt} + 2y = 1 \equiv \frac{dy}{dx} + P(x)y = Q(x)$$

$$\therefore P(x) = 2$$

$$Q(x) = 1$$

$$\Rightarrow u(x) = e^{\int P(x) dt} = e^{\int 2 dt} = e^{2t}$$

$$\therefore e^{2t} \left[\frac{dy}{dt} + 2y = 1 \right] \Rightarrow e^{2t} \frac{dy}{dt} + e^{2t} 2y = e^{2t}$$

$$e^{2t} \frac{dy}{dt} + e^{2t} 2y = e^{2t} \Rightarrow \int \frac{d}{dt} [e^{2t} y] dt = \int e^{2t} dt$$

$$\Rightarrow \frac{d}{dt} [e^{2t} y] = e^{2t} \Rightarrow (e^{2t} y = \frac{1}{2} e^{2t} + C) \times 2$$

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$$\Rightarrow 2y e^{2t} = e^{2t} + 2C \rightarrow \text{let this be another constant, } C_1$$

$$\Rightarrow 2y e^{2t} = e^{2t} + C_1$$

$$\Rightarrow y = \frac{e^{2t} + C_1}{2e^{2t}}$$

Given the initial-value; $y = 5/2$, $t = 0$; $y \neq 1/2$

$$\therefore \frac{5}{2} = \frac{e^0 + C_1}{2e^0}; \frac{5}{2} = \frac{1 + C_1}{2 \times 2}$$

$$\Rightarrow 5 = 1 + C_1$$

$$5 - 1 = C_1$$

$$\Rightarrow C_1 = 4$$

$$\therefore y = \frac{e^{2t} + 4}{2e^{2t}} = \frac{e^{2t}}{2e^{2t}} + \frac{4}{2e^{2t}}$$

$$y = \frac{2}{e^{2t}} + \frac{1}{2}; y = 2e^{-2t} + \frac{1}{2}$$

Q27.

$$\sqrt{1-y^2} dx - \sqrt{1-x^2} dy = 0$$

$$\sqrt{1-y^2} dx = \sqrt{1-x^2} dy$$

$$\frac{1}{\sqrt{1-x^2}} dx = \frac{1}{\sqrt{1-y^2}} dy \quad \left\{ \int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C \right.$$

$$\therefore \sin^{-1}(x) = \sin^{-1}(y) + C$$

Given the initial-value; $y = \frac{\sqrt{3}}{2}$, $x = 0$

$$\Rightarrow \sin^{-1}(0) = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) + C \quad \therefore \sin^{-1}(x) = \sin^{-1}(y) + C$$

$$0 = \frac{\pi}{3} + C$$

$$\Rightarrow x = y + \left(\sin\left(-\frac{\pi}{3}\right)\right)$$

$$-\frac{\pi}{3} = C$$

$$x = y - \sin\left(\frac{\pi}{3}\right)$$

$$\Rightarrow -y = -x - \frac{\sqrt{3}}{2}; -y = -1\left(x + \frac{\sqrt{3}}{2}\right)$$

$$\Rightarrow y = x + \frac{\sqrt{3}}{2}$$

Q28. $(1+x^4)dy + x(1+4y^2)dx = 0$
 $(1+x^4)dy = -x(1+4y^2)dx$
 $\Rightarrow \frac{1}{1+4y^2} dy = \frac{-x}{1+x^4} dx$

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$$\frac{1}{1+4y^2} = \frac{1}{1+(2y)^2} dy \Rightarrow \int \frac{1}{1+x^2} dx = \tan^{-1}(x)$$

$$\therefore \int \frac{1}{1+4y^2} dy = \int \frac{-x}{1+x^4} dx$$

$$\Rightarrow \tan^{-1}(2y) = \int \frac{-x}{1+x^4} dx$$

$$\Rightarrow \text{Let } u = x^2 ; du = 2x dx ; \frac{1}{2} du = x dx$$

$$\Rightarrow \int \frac{-x}{1+x^4} \left(\frac{1}{2x} \right) du \Rightarrow -\frac{1}{2} \int \frac{1}{u^2+1} du$$

$$\Rightarrow \frac{-1}{2} \tan^{-1}(u) \Rightarrow \frac{-1}{2} \tan^{-1}(x^2)$$

$$\therefore \tan^{-1}(2y) = \frac{-1}{2} \tan^{-1}(x^2) + C$$

$$\Rightarrow 2y = \frac{-1}{2} (x^2) + \tan C$$

$$y = \frac{-1}{4} (x^2) + \frac{\tan C}{2}$$

Given the initial value: $y=0$, $x=1$;

$$\Rightarrow 0 = \frac{-1}{4} (1) + \frac{\tan C}{2}$$

$$\left(0 = \frac{-1}{4} + \frac{\tan C}{2} \right) \times 4 \Rightarrow 0 = -1 + 2 \tan C$$

$$1 = 2 \tan C$$

$$\frac{1}{2} = \tan C$$

$$\tan^{-1}(1/2) = C$$

$$\therefore y = \frac{-x^2}{4} + \frac{\tan(\tan^{-1}(1/2))}{2}$$

$$y = \frac{-x^2}{4} + \frac{1}{4}$$

$$y = \frac{-x^2+1}{4}$$

Q1. $\frac{dy}{dx} = 5y$; $dy = 5y dx$

$$\Rightarrow \int \frac{1}{5y} dy = \int dx \Rightarrow \frac{1}{5} \int \frac{1}{y} dy = x + C$$

$$\Rightarrow \frac{1}{5} \ln y = x + C \Rightarrow \ln y = 5x + 5C$$

$$\Rightarrow \log_e y = 5x + 5C ; y = e^{5x+5C}$$

$$\Rightarrow y = e^{5x+5C} = e^{5x} \cdot e^{5C}$$

$$= e^{5x} \cdot C_1$$

$$\Rightarrow \therefore \underline{\underline{y = C_1 e^{5x}}}$$

Largest Interval : solution defined for all real x , so, the largest interval is $(-\infty, \infty)$.

Transient Terms : No transient terms since the exponential term e^{5x} does not decay to zero.

Q2. $\frac{dy}{dx} + 2y = 0$; $\frac{dy}{dx} = -2y$; $dy = -2y dx$

$$\Rightarrow \int \frac{-1}{2y} dy = \int dx \Rightarrow \frac{-1}{2} \ln y = x + C$$

$$\Rightarrow -\ln y = 2x + 2C$$

$$\Rightarrow \ln y = -2x - 2C$$

$$\log_e y = -2x - 2C$$

$$y = e^{-2x-2C}$$

$$y = e^{-2x} \cdot e^{-2C}$$

$$y = e^{-2x} C_1$$

$$\underline{\underline{y = C_1 e^{-2x}}}$$

Largest Interval : defined for all x , in the interval $(-\infty, \infty)$.

Transient Terms : e^{-2x} since it decays to zero as $x \rightarrow \infty$.

Q3. $\frac{dy}{dx} + y = e^{3x}$
 $\Rightarrow \frac{dy}{dx} + P(x)y = Q(x)$ } first-order linear differential equation

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$$\therefore \mu = e^{\int P(x) dx} = e^{\int 1 dx} = e^x$$

$$\Rightarrow \frac{dy}{dx} e^x + e^x y = e^{3x} \cdot e^x$$

$$\Rightarrow \frac{d}{dx} [e^x \cdot y] = e^{3x+x} = e^{4x}$$

$$\Rightarrow \int \frac{d}{dx} [e^x y] = \int e^{4x}$$

$$\Rightarrow e^x y = \frac{1}{4} e^{4x} + C$$

$$\Rightarrow y = \frac{e^{4x-x}}{4} + \frac{C}{e^x}$$

$$y = \frac{e^{3x}}{4} + \frac{C}{e^x}$$

Q4. $3 \frac{dy}{dx} + 12y = 4$; $\frac{dy}{dx} + 4y = \frac{4}{3}$

$$\Rightarrow \frac{dy}{dx} + P(x)y = Q(x)$$

$$\Rightarrow \mu = e^{\int P(x) dx} = e^{\int 4 dx} = e^{4x}$$

$$\Rightarrow e^{4x} \frac{dy}{dx} + e^{4x} 4y = \frac{4e^{4x}}{3}$$

$$\Rightarrow \int \frac{d}{dx} [e^{4x} \cdot y] = \int \frac{4}{3} e^{4x}$$

$$e^{4x} \cdot y = \frac{4}{3} \left(\frac{1}{4} \right) e^{4x} + C$$

$$e^{4x} y = \frac{4}{12} e^{4x} + C$$

$$e^{4x} y = \frac{e^{4x}}{3} + C$$

$$y = \frac{e^{4x}}{3 e^{4x}} + \frac{C}{e^{4x}}$$

$$y = \frac{C}{e^{4x}} + \frac{1}{3}$$

Q5. $y' + 3x^2y = x^2$ $\Rightarrow u = e^{\int p(x)dx}$
 $\Rightarrow \frac{dy}{dx} + 3x^2y = x^2$ $= e^{\int 3x^2dx}$
 $\Rightarrow \frac{dy}{dx} + p(x)y = Q(x)$ $= e^{3(\frac{x^3}{3})}$
 $= e^{\frac{3x^3}{3}} = e^{x^3}$

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$$\Rightarrow e^{x^3} \left(\frac{dy}{dx} + 3x^2y = x^2 \right)$$

$$\Rightarrow e^{x^3} \frac{dy}{dx} + 3x^2y e^{x^3} = x^2 e^{x^3}$$

$$\Rightarrow \int \frac{d}{dx} [e^{x^3} \cdot y] = \int x^2 e^{x^3}$$

$$e^{x^3} \cdot y = \frac{1}{3} e^{x^3} + C$$

$$y = \frac{e^{x^3}}{3e^{x^3}} + \frac{C}{e^{x^3}}$$

$$y = \frac{C}{e^{x^3}} + \frac{1}{3}$$

Q6. $y' + 2xy = x^3$ $u = e^{\int p(x)dx}$
 $\Rightarrow \frac{dy}{dx} + 2xy = x^3$ $= e^{\int 2x dx}$
 $\Rightarrow \frac{dy}{dx} + p(x)y = Q(x)$ $= e^{\frac{2x^2}{2}} = e^{x^2}$

$$\Rightarrow e^{x^2} \left(\frac{dy}{dx} + 2xy e^{x^2} = x^3 e^{x^2} \right)$$

$$\Rightarrow e^{x^2} \frac{dy}{dx} + 2xy e^{x^2} = x^3 e^{x^2}$$

$$\Rightarrow [e^{x^2} \cdot y] \frac{d}{dx} = x^3 e^{x^2}$$

$$\Rightarrow \int \frac{d}{dx} [e^{x^2} \cdot y] = \int x^3 e^{x^2}$$

$$\Rightarrow e^{x^2} \cdot y = \int x^3 e^{x^2} dx \quad \text{Let } u = x^2$$

$$\Rightarrow e^{x^2} y = \int \frac{x^2 e^u}{2x} du$$

$$\frac{du}{dx} = 2x$$

$$dx = \frac{1}{2x} du$$

$$\Rightarrow e^{x^2} y = \frac{1}{2} \int x^2 e^u du = \frac{1}{2} \int u e^u du \rightarrow \text{integration by parts}$$

$$\Rightarrow e^{x^2} y = \frac{1}{2} \left[u \int v dx - \left[\frac{du}{dx} \times \int v dx \right] dx \right]$$

$$\Rightarrow e^{x^2} y = \frac{1}{2} \left[u e^u - \int [1 \times e^u] du \right] = \frac{1}{2} [u e^u - e^u]$$

$$\Rightarrow e^{x^2} y = \frac{1}{2} [u e^u - e^u] = \frac{1}{2} e^u (u-1)$$

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$$e^{x^2} y = \frac{1}{2} e^u (u-1)$$

$$e^{x^2} y = \frac{1}{2} e^{x^2} (x^2-1) + C$$

$$y = \frac{e^{x^2} (x^2-1)}{2 e^{x^2}} + \frac{C}{e^{x^2}}$$

$$y = \frac{x^2-1}{2} + \frac{C}{e^{x^2}}$$

Q7.

$$x^2 y' + xy = 1$$

$$\Rightarrow \frac{dy}{dx} + \frac{xy}{x^2} = \frac{1}{x^2}$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = \frac{1}{x^2}$$

$$\Rightarrow \frac{dy}{dx} + P(x)y = Q(x)$$

$$\Rightarrow \left(\frac{dy}{dx} + \frac{1}{x} (y) = \frac{1}{x^2} \right) \times x$$

$$\Rightarrow x \frac{dy}{dx} + y = \frac{1}{x}$$

$$\Rightarrow \int \frac{d}{dx} [xy] = \int \frac{1}{x} ; xy = \ln x + C$$

$$\Rightarrow y = \frac{\ln x}{x} + \frac{C}{x}$$

Largest Interval : $\ln x$ is only defined for $x > 0$
and $1/x$ is undefined at $x=0$.
 $\therefore (0, \infty)$

Transient Terms : Both terms are transient,
since they both decay to
zero, as $x \rightarrow \infty$.

Q8.

$$y' = 2y + x^2 + 5$$

$$\Rightarrow y' - 2y = x^2 + 5$$

$$\Rightarrow \frac{dy}{dx} + p(x)y = Q(x)$$

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$$e^{\int p(x)dx} = u$$

$$e^{\int -2dx} = e^{-2x}$$

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$$\Rightarrow e^{-2x} \frac{dy}{dx} - 2ye^{-2x} = e^{-2x}x^2 + 5e^{-2x}$$

$$\Rightarrow \int \frac{d}{dx} [e^{-2x}y] = \int e^{-2x}x^2 + 5e^{-2x}$$

$$e^{-2x}y = \underbrace{\int e^{-2x}x^2 dx} + \int 5e^{-2x} dx$$

Integration by parts:

$$u \int v dx - \int [u' \times \int v dx] dx$$

$$x^2 \left(-\frac{1}{2}\right) e^{-2x} - \int 2x \times \left(-\frac{e^{-2x}}{2}\right) dx$$

$$\frac{-x^2 e^{-2x}}{2} - \int x e^{-2x} dx \Rightarrow \frac{-x^2 e^{-2x}}{2} - \left[x \left(-\frac{e^{-2x}}{2}\right) - \int \frac{-e^{-2x}}{2} dx \right]$$

$$\frac{-x^2 e^{-2x}}{2} - \left[\frac{-x e^{-2x}}{2} + \frac{1}{2} \left(-\frac{1}{2}\right) e^{-2x} \right]$$

$$\frac{-x^2 e^{-2x}}{2} + \frac{x e^{-2x}}{2} + \frac{e^{-2x}}{4}$$

$$e^{-2x} \left(\frac{-x^2}{2} + \frac{x}{2} + \frac{1}{4} \right)$$

$$\Rightarrow e^{-2x}y = e^{-2x} \left(\frac{-x^2}{2} + \frac{x}{2} + \frac{1}{4} \right) + 5 \left(-\frac{1}{2}\right) e^{-2x} + C$$

$$\Rightarrow e^{-2x}y = e^{-2x} \left(\frac{-x^2}{2} + \frac{x}{2} + \frac{1}{4} \right) - \frac{5e^{-2x}}{2} + C$$

$$y = \left(\frac{-x^2}{2} + \frac{x}{2} + \frac{1}{4} \right) - \frac{5}{2} + \frac{C}{e^{-2x}}$$

$$y = \frac{-x^2}{2} + \frac{x}{2} - \frac{9}{4} + C e^{2x}$$

Largest Interval: defined for all x ; $(-\infty, \infty)$

Transient terms: NO transient terms;

all terms either grow or stay constant as $x \rightarrow \infty$.

Q9. $x \frac{dy}{dx} - y = x^2 \sin x$

$$\frac{dy}{dx} - \frac{y}{x} = \frac{x^2 \sin x}{x}$$

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$$\frac{dy}{dx} - \frac{1}{x}(y) = x \sin x$$

$$\frac{dy}{dx} - \frac{1}{x}y = x \sin x$$

$$\Rightarrow \frac{dy}{dx} + P(x)y = Q(x)$$

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$$\Rightarrow \mu = e^{\int P(x) dx}$$

$$= e^{\int -1/x dx} = e^{-\ln x} = e^{(-1)(\ln x)} = e^{\ln x^{-1}}$$

$$= e^{\ln(1/x)} = \frac{1}{x}$$

$$\Rightarrow \frac{1}{x} \frac{dy}{dx} - \frac{1}{x^2}(y) = \sin x$$

$$\Rightarrow \int \frac{d}{dx} \left[\frac{y}{x} \right] = \int \sin x$$

$$\frac{y}{x} = -\cos x + C$$

$$y = -x \cos x + xC$$

Q10.

$$\frac{dy}{dx} + \frac{2y}{x} = \frac{3}{x}$$

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$\Rightarrow \mu = e^{\int 2/x dx} = e^{2 \ln x} = e^{\ln x^2} = x^2$$

$$\Rightarrow x^2 \frac{dy}{dx} + \frac{2y}{x}(x^2) = \frac{3}{x}(x^2)$$

$$\Rightarrow x^2 \frac{dy}{dx} + 2xy = 3x$$

$$\Rightarrow \int \frac{d}{dx} [x^2 y] = \int 3x$$

$$\Rightarrow x^2 y = \frac{3x^2}{2} + C$$

$$\Rightarrow y = \frac{3x^2}{2x^2} + \frac{C}{x^2}$$

$$\Rightarrow y = \frac{3}{2} + \frac{C}{x^2}$$

Q11. $\frac{dy}{dx} + \frac{4y}{x} = x^2 - 1$

$\Rightarrow \frac{dy}{dx} + P(x)y = Q(x)$

$\mu = e^{\int P(x)dx}$

$= e^{\int 4/x dx}$

$= e^{4 \ln x} = e^{\ln x^4} = x^4$

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$\Rightarrow x^4 \frac{dy}{dx} + \frac{4y}{x} (x^4) = (x^2 - 1)x^4$

$\Rightarrow x^4 \frac{dy}{dx} + 4yx^3 = x^6 - x^4$

$\Rightarrow \int \frac{d}{dx} [x^4 y] = \int x^6 - x^4$

$x^4 y = \frac{x^7}{7} - \frac{x^5}{5} + C$

$y = \frac{x^7}{7x^4} - \frac{x^5}{5x^4} + \frac{C}{x^4}$

$y = \frac{x^3}{7} - \frac{x}{5} + \frac{C}{x^4}$

$y = \frac{x^3}{7} - \frac{x}{5} + \frac{C}{x^4}$

Q12.

~~$(1+x) \frac{dy}{dx} + \frac{y}{1+x} = x + \frac{1}{1+x}$~~

~~$\Rightarrow \frac{d}{dx} \left[\frac{y}{1+x} \right] = \frac{x+1}{1+x}$~~

~~$\Rightarrow \frac{d}{dx} \left[\frac{y}{1+x} \right] = 1$~~

~~$\Rightarrow \frac{y}{1+x} = x + C$~~

~~$y = (x+C)(1+x)$~~

~~$y = x^2 + Cx + x + C$~~

~~$y = x^2 + (C+1)x + C$~~

~~$\Rightarrow \mu = e^{\int P(x)dx} = e^{\int \frac{1}{1+x} dx} = e^{\ln(1+x)} = 1+x$~~

~~$\Rightarrow \mu y = \int Q(x)\mu dx$~~

~~$(1+x)y = \int (x+1) dx$~~

~~$(1+x)y = \frac{x^2}{2} + x + C$~~

~~$y = \frac{x^2}{2(1+x)} + \frac{x}{1+x} + \frac{C}{1+x}$~~

Q12. $\frac{dy}{dx} + \left(-\frac{xy}{1+x}\right) = \frac{x-1x^2}{1+x}$; $\frac{dy}{dx} - \frac{xy}{1+x} = x$
 $\Rightarrow \frac{dy}{dx} - \frac{xy}{1+x} = \frac{x(1+x)}{1+x}$; $\Rightarrow \frac{dy}{dx} + P(x)y = Q(x)$

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$\Rightarrow e^{\int P(x)dx} = u \Rightarrow \int -\frac{x}{1+x} dx = - \int \frac{x}{1+x} dx$; let $u = 1+x$
 $dx = du$

$\Rightarrow - \int \frac{u-1}{u} du = - \left[\int 1 du - \int \frac{1}{u} du \right]$ $u-1 = x$

$\Rightarrow - \left[u - \ln u \right] = -u + \ln u = \ln u - u = \ln(1+x) - (1+x)$
 $= \ln(1+x) - 1 - x$
 $= \ln(x+1) - (1+x)$

$\Rightarrow e^{\ln(x+1)} \cdot e^{-(1+x)} = (x+1)e^{-(1+x)}$

$\Rightarrow (x+1)e^{-(1+x)} \frac{dy}{dx} - \frac{xy}{1+x} e^{-(1+x)} = (x+1)e^{-(1+x)} (x)$

$\Rightarrow \frac{d}{dx} [ye^{-x(1+x)}] = e^{-x(1+x)} x$

$ye^{-x(1+x)} = \int e^{-x(1+x)} x$

$ye^{-x(1+x)} = -x^2 e^{-x} - 3x e^{-x} - 3e^{-x} + C$

$\Rightarrow y = \frac{-x^2 - 3x - 3}{1+x} + \frac{Ce^x}{1+x}$

Q13. $x^2 y' + x(x+2)y = e^x$
 $x^2 \frac{dy}{dx} + (x^2 + 2x)y = e^x$

$\frac{dy}{dx} + \frac{x(x+2)}{x^2} y = \frac{e^x}{x^2}$

$\frac{dy}{dx} + \frac{(x+2)}{x} y = \frac{e^x}{x^2}$

$\Rightarrow \frac{dy}{dx} + P(x)y = Q(x)$

$\Rightarrow u = e^{\int P(x)dx} = e^{\int \frac{1}{x} + \frac{2}{x} dx}$

$\Rightarrow e^{x+2\ln x} = e^{x+1\ln x^2} = e^x \cdot e^{\ln x^2}$

$= e^x \cdot x^2$

$\therefore e^x x^2 \frac{dy}{dx} + \frac{(x+2)y}{x} (e^x x^2) = e^x e^x x^2$

$\Rightarrow e^x x^2 \frac{dy}{dx} + (x+2)(y)(e^x x) = e^{2x}$

$\Rightarrow \int \frac{d}{dx} [e^x x^2 y] = \int e^{2x}$

$e^x x^2 y = \frac{1}{2} e^{2x} + C$

$y = \frac{e^{2x}}{2e^x x^2} + \frac{C}{e^x x^2}$

$y = \frac{e^{2x-x}}{2x^2} + \frac{C}{e^x x^2}$

$y = \frac{e^x}{2x^2} + \frac{C}{e^x x^2}$

Q14. $xy' + (1+x)y = e^{-x} \sin 2x$ | $\mu = e^{\int 1/x + 1 dx}$
 $= e^{\ln x + x} = e^{\ln x} \cdot e^x = x e^x$
 $\frac{dy}{dx} + (1+x)y = e^{-x} \sin 2x$
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$$\Rightarrow \frac{dy}{dx} + P(x)y = Q(x)$$

$$\therefore x e^x \frac{dy}{dx} + \frac{(1+x)(y)}{x} (x e^x) = \frac{e^{-x} \sin 2x}{x} (x e^x)$$

$$x e^x \frac{dy}{dx} + (1+x)(y)(e^x) = \sin 2x$$

$$\Rightarrow \int \frac{d}{dx} [x e^x y] = \int \sin 2x$$

$$\Rightarrow x e^x y = \frac{1}{2} (-\cos 2x) + C$$

$$y = \frac{-\cos 2x}{2x e^x} + \frac{C}{x e^x}$$

Q15. $y dx - 4(x+y^6) dy = 0$; $M(x,y) = y$

for an exact differential eq. ; $N(x,y) = -4x - 4y^6$

$\Rightarrow M_y = 1$; Since $M_y \neq N_x$; it is not

$N_x = -4 - 4y^6 = -4$ an exact equation.

$$y dx - 4(x+y^6) dy = 0$$

$$y dx = 4(x+y^6) dy$$

$$\frac{dx}{dy} = \frac{4(x+y^6)}{y} = \frac{4x}{y} + 4y^5$$

$\frac{dx}{dy} = \frac{4x}{y} + 4y^5$ } Linear ODE in x with respect to y ;

$\Rightarrow \frac{dx}{dy} - \frac{4x}{y} = 4y^5$ } $\Rightarrow \frac{dx}{dy} + P(y)x = Q(y)$

$\Rightarrow P(y) = -\frac{4}{y}$; $\mu = e^{\int -4/y dy} = e^{-4 \ln y} = e^{\ln y^{-4}} = y^{-4}$

$\Rightarrow y^4 \frac{dx}{dy} - \frac{4x(y^{-4})}{y} = \frac{4y^5}{y^4}$ | $\frac{1}{y^4} \frac{dx}{dy} - \frac{4x}{y^5} = 4y$

$\Rightarrow y^4 \frac{dx}{dy} - \frac{4x}{y^5} = 4y$ | $\int \frac{d}{dy} [x y^{-4}] = \int 4y$

$$\Rightarrow xy^{-4} = 2y^2 + C$$

$$\Rightarrow x = y^4(2y^2 + C) = 2y^{-2} + Cy^{-4}$$

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$$\Rightarrow x = \frac{2}{y^2} + \frac{C}{y^4}$$

Q21. $(x+y)^2 dx + (2xy+x^2-1)dy = 0$

$$\Rightarrow M(x,y) = (x+y)^2 = x^2 + 2xy + y^2$$

$$N(x,y) = 2xy + x^2 - 1$$

$\Rightarrow$ In order to test for an exact differential eq.,

$$\Rightarrow M_y = N_x$$

$$\Rightarrow 2x + 2y = 2y + 2x \quad \text{Q.E.D}$$

$$\frac{\partial \Psi}{\partial x} = M \quad ; \quad \frac{\partial \Psi}{\partial y} = N$$

$$\int \frac{\partial \Psi}{\partial x} = M = \int x^2 + 2xy + y^2$$

$$\Rightarrow \int \Psi_x dx = \int x^2 + 2xy + y^2 dx$$

$$\Rightarrow \Psi = \left[\frac{x^3}{3} + \frac{2x^2y}{2} \right] + f(y)$$

$$\Rightarrow \frac{\partial \Psi}{\partial y} = 0 + \frac{2x^2}{2} + f'(y)$$

$$\frac{\partial \Psi}{\partial y} = x^2 + f'(y)$$

$$\Rightarrow x^2 + f'(y) = 2xy + x^2 - 1$$

$$f'(y) = 2xy + x^2 - 1 - x^2$$

$$f'(y) = 2xy - 1$$

Q22. $(e^x + y) \frac{dx}{M(x,y)} + (2 + x + ye^y) \frac{dy}{N(x,y)} = 0$

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$$\frac{\partial F}{\partial x} = M(x,y) = e^x + y$$

$$\frac{\partial F}{\partial y} = N(x,y) = 2 + x + ye^y$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow 1 = 1 \text{ Q.E.D.}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the equation is an exact equation.

$$\Rightarrow \int \frac{\partial F}{\partial x} = \int (e^x + y) dx \Rightarrow F(x,y) = e^x + xy + g(y)$$

$$\Rightarrow \frac{\partial F}{\partial y} = 0 + x + g'(y) = 2 + x + g'(y)$$

$$\Rightarrow x + g'(y) = 2 + x + ye^y$$

$$g'(y) = 2 + x - x + ye^y$$

$$\int g'(y) = \int 2 + ye^y$$

$$g(y) = 2y + \left[ye^y - \int 1 \times e^y dy \right] = 2y + [ye^y - e^y]$$

$$g(y) = 2y + [e^y(y-1)] = 2y + e^y(y-1)$$

$$g(y) = 2y + e^y(y-1)$$

$$\Rightarrow \underline{e^x + xy + 2y + e^y(y-1) = c}$$

$$\Rightarrow y=1, \quad x=0$$

$$\Rightarrow e^0 + 0 + 2(1) + e(0) = c$$

$$\Rightarrow 1 + 0 + 2 + 0 = c$$

$$\Rightarrow 3 = c$$

$$\therefore \underline{e^x + xy + 2y + e^y(y-1) = 3}$$

$$\text{Q23. } \underbrace{(4y+2t-5)}_{M(t,y)} dt + \underbrace{(6y+4t-1)}_{N(t,y)} dy = 0$$

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$$\frac{\partial F}{\partial x} = M(t,y) = 4y+2t-5$$

$\frac{\partial F}{\partial x}$

$$\frac{\partial F}{\partial y} = N(t,y) = 6y+4t-1$$

$\frac{\partial F}{\partial y}$

$$\Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial t} \quad ; \quad 4 = 4 \quad ; \quad \text{O.E.D.}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}$; the equation is exact.

$$\int \frac{\partial F}{\partial x} = \int (4y+2t-5) dx : F(t,y) = 4xy + \frac{2t^2}{2} - 5t + g(y)$$

$$F(t,y) = 4xy + \frac{2t^2}{2} - 5t + g(y) = 4xy + t^2 - 5t + g(y)$$

$$\frac{\partial F}{\partial y} = 4t + g'(y)$$

$\frac{\partial F}{\partial y}$

$$\Rightarrow 4t + g'(y) = 6y + 4t - 1$$

$$\int g'(y) = \int 6y - 1$$

$$g(y) = \frac{6y^2}{2} - y = 3y^2 - y$$

$$\Rightarrow f(t,y) = 4xy + t^2 - 5t + 3y^2 - y = c$$

$$\Rightarrow y=2, \quad x=-1$$

$$\Rightarrow 4(-1)(2) + (-1)^2 - 5(-1) + 12 - 2 = c$$

$$-8 + 1 + 5 + 10 = c$$

$$c = 8$$

$$\Rightarrow 4xy + t^2 - 5t + 3y^2 - y = 8$$

$$\Rightarrow \underline{4xy + t^2 - 5t + 3y^2 - y = 8}$$

$$Q24. \left(\frac{3y^2 - t^2}{y^5} \right) dy + \frac{t}{2y^4} dt = 0 \quad \left| \frac{t}{2y^4} dt + \left(\frac{3y^2 - t^2}{y^5} \right) dy = 0 \right.$$

$$\Rightarrow \left(\frac{3y^2 - t^2}{y^5} \right) dy + \frac{t}{2y^4} dt = 0$$

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$$\Rightarrow \frac{\partial F}{\partial t} = \frac{t}{2y^4} = M(t, y)$$

$$\frac{\partial F}{\partial y} = \frac{3y^2 - t^2}{y^5} = N(t, y)$$

$$\Rightarrow \frac{\partial M}{\partial y} = \frac{t}{2} [-4y^{-5}] = \frac{-4t^2}{2y^5} = \frac{-2t}{y^5}$$

$$\Rightarrow \frac{\partial N}{\partial t} = 0 - 1(2t) = \frac{-2t}{y^5}$$

$\Rightarrow \frac{-2t}{y^5} = \frac{\partial M}{\partial y} = \frac{-2t}{y^5} = \frac{\partial N}{\partial t}$; since both the partial derivatives are equal, the equation is an exact differential equation.

$$\Rightarrow \int \frac{\partial F}{\partial t} = \int \frac{t}{2y^4} dt = \frac{1}{2y^4} \left[\frac{t^2}{2} \right] = \frac{t^2}{4y^4} + g(y)$$

$$\frac{\partial F}{\partial y} = \frac{t^2}{4} [-4y^{-5}] + g'(y) = \frac{-4t^2}{4y^5} + g'(y)$$

$$\Rightarrow \frac{-4t^2}{4y^5} + g'(y) = \frac{3y^2 - t^2}{y^5}$$

$$\frac{-t^2}{y^5} + g'(y) = \frac{3y^2 - t^2}{y^5}$$

$$\frac{-t^2}{y^5} + g'(y) = \frac{3y^2}{y^5} - \frac{t^2}{y^5}$$

$$g'(y) = \frac{3}{y^3} - \frac{t^2}{y^5} + \frac{t^2}{y^5}$$

$$\int g'(y) = \int \frac{3}{y^3}$$

$$g(y) = 3 \int \frac{1}{y^3} dy = 3 \int y^{-3} dy = 3 \left[\frac{y^{-2}}{-2} \right]$$

$$g(y) = -\frac{3}{2y^2}$$

$$\Rightarrow F(t, y) = \frac{t^2}{4y^4} - \frac{3}{2y^2} = c$$

$$\frac{1}{4y^4} - \frac{3}{2y^2} = C$$

$$y=1, x=1$$

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$$\therefore \frac{1}{4} - \frac{3}{2} = C; \quad \frac{1}{4} - \frac{6}{4} = C = -\frac{5}{4}$$

$$\Rightarrow \frac{1^2}{4y^4} - \frac{3}{2y^2} = -\frac{5}{4}$$

825. $(y^2 \cos x - 3x^2 y - 2x) dx + (2y \sin x - x^3 + \ln y) dy = 0$

$$\frac{\partial F}{\partial x} = M(x, y) = y^2 \cos x - 3x^2 y - 2x$$

$$\frac{\partial F}{\partial y} = N(x, y) = 2y \sin x - x^3 + \ln y$$

$$\frac{\partial M}{\partial y} = 2 \cos x y - 3x^2; \quad \frac{\partial N}{\partial x} = 2y \cos x - 3x^2$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$; the equation is an exact differential equation.

$$\int \frac{\partial F}{\partial x} = \int (y^2 \cos x - 3x^2 y - 2x) dx$$

$$F(x, y) = \cancel{\cos x \left[\frac{y^3}{3} \right]} - \cancel{3x^2 \left[\frac{y^2}{2} \right]} + y^2 (\sin x) - \frac{3x^3}{3} (y) - \frac{2x^2}{2}$$

$$F(x, y) = y^2 \sin x - x^3 y - x^2 + g(y)$$

$$\frac{\partial F}{\partial y} = 2 \sin x y - x^3 + g'(y)$$

$$\Rightarrow 2 \sin x y - x^3 + g'(y) = 2y \sin x - x^3 + \ln y$$

$$\int g'(y) = \int \ln y$$

$$g(y) = \int \ln y dy = \int \left[\frac{du}{u} \times \int u dy \right] dy$$

$$= \ln y (y) - \int \frac{1}{y} \times y dy = \ln y (y) - y = y(\ln y - 1)$$

$$\Rightarrow f(x, y) = y^2 \sin x - x^3 y - x^2 + y(\ln y - 1) = C \quad \therefore y^2 \sin x - x^3 y - x^2 + y(\ln y - 1) = 0$$

$$\Rightarrow y = e, x = 0$$

$$\Rightarrow e^2 \sin 0 - 0 - 0 + e(1 - 1) = C; \quad C = 0$$

$$26. \left(\frac{1}{1+y^2} + \cos x - 2xy \right) \frac{dy}{dx} = y(y + \sin x)$$

$$\left(\frac{1}{1+y^2} + \cos x - 2xy \right) dy = (y^2 + y \sin x) dx$$

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$$\left(\frac{1}{1+y^2} + \cos x - 2xy \right) dy - (y^2 + y \sin x) dx = 0$$

$$(y^2 + y \sin x) dx - \left(\frac{1}{1+y^2} + \cos x - 2xy \right) dy = 0$$

$$(y^2 + y \sin x) dx - (-1) \left(2xy - \frac{1}{1+y^2} - \cos x \right) dy = 0$$

$$(y^2 + y \sin x) dx + \left(2xy - \frac{1}{1+y^2} - \cos x \right) dy = 0 \quad \left. \vphantom{\frac{1}{1+y^2}} \right\} \text{brought to standard form}$$

$$\frac{\partial F}{\partial x} = y^2 + y \sin x = M(x, y)$$

$$\frac{\partial F}{\partial y} = 2xy - \frac{1}{1+y^2} - \cos x = N(x, y)$$

$$\Rightarrow \frac{\partial M}{\partial y} = 2y + \sin x ; \frac{\partial N}{\partial x} = 2y - (-\sin x) = 2y + \sin x$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the equation is an exact differential eq.

$$\int \frac{\partial F}{\partial x} = \int y^2 + y \sin x \, dx = f(x, y) = xy^2 + y(-\cos x) + g(y)$$

$$\Rightarrow f(x, y) = xy^2 - y \cos x + g(y)$$

$$\Rightarrow \frac{\partial F}{\partial y} = x(2y) - \cos x + g'(y) = 2xy - \cos x + g'(y)$$

$$\Rightarrow 2xy - \cos x + g'(y) = 2xy - \frac{1}{1+y^2} - \cos x$$

$$\int g'(y) = \int -\frac{1}{1+y^2} = -\tan^{-1}(y) = g(y)$$

$$\Rightarrow F(x, y) = xy^2 - y \cos x - \tan^{-1}(y) = C$$

$$\Rightarrow y=1, x=0$$

$$\Rightarrow 0 - 1 \cos 0 - \tan^{-1}(1) = C$$

$$\Rightarrow 0 - 1 - \frac{\pi}{4} = C ; C = -1 - \frac{\pi}{4} = \frac{-4-\pi}{4}$$

$$\Rightarrow \nexists xy^2 - y \cos x - \tan^{-1}(y) = \frac{-4-\pi}{4}$$